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EARTH SYSTEM SCIENCES



Role of Livestock Enteric Methane in Climate Vulnerability in India (2003 – 2022) (Poster #3199)

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Photo: pixabay/RonaldPlett

To examine the role and impact of cattle enteric methane on climate vulnerability.

Livestock contributes to 35–40% of methane emissions globally.

Agriculture is the predominant source of methane largely originating from livestock.

49.1% of enteric methane was contributed by cattle, 42.8% by buffaloes, 5.38% by goat and 2.59% by sheep[1]. The beef and dairy cattle industry is responsible for generating 14.5% of global greenhouse gas emissions[2]. During 1961–2010, the increase in CH₄ emissions (70.6%) from livestock population of India is much greater than the increase in CH₄ emissions from livestock population of world (54.3%)[1].

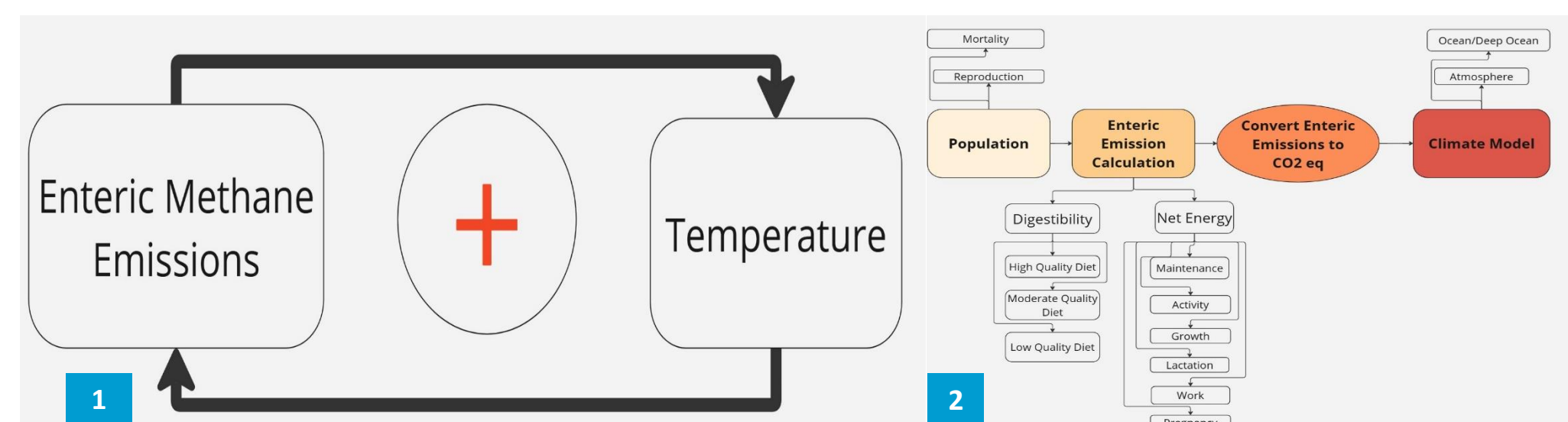
Cattle emit CH₄ through burping and flatulence[3]. Enteric fermentation occurs in the cow's rumen (first chamber of the stomach), where microbes break down feed, producing volatile fatty acids (VFAs) and gases like methane. VFAs provide energy for growth and milk production, while CH₄ is released through belching. Diet, feeding patterns, and physiological state influence VFA production and methane emissions.[4]

Different cattle breeds emit different quantities of CH₄ in different physiological stages. During 2014 – 15 and 2021 – 22, the CH₄ emissions from Indigenous Cattle grew by 14.25% and from Crossbred Cattle by 59.94%.

Climate change will affect livestock production significantly through their effects on feed production feed quality and composition, pest and disease dynamics, and animal health and production. Extreme events like droughts (Drought of 1987) affected 168 million cattle due to feed and fodder availability and water scarcity[1].

Methane is a potent GHG with 23 times the warming potential of Carbon dioxide.

CH₄ emissions and temperature creates a bidirectional feedback loop; a self-reinforcing cycle in which increased temperatures stimulate CH₄ and vice-versa.



- 1) Bidirectional feedback loop representation.
- 2) Model representation.

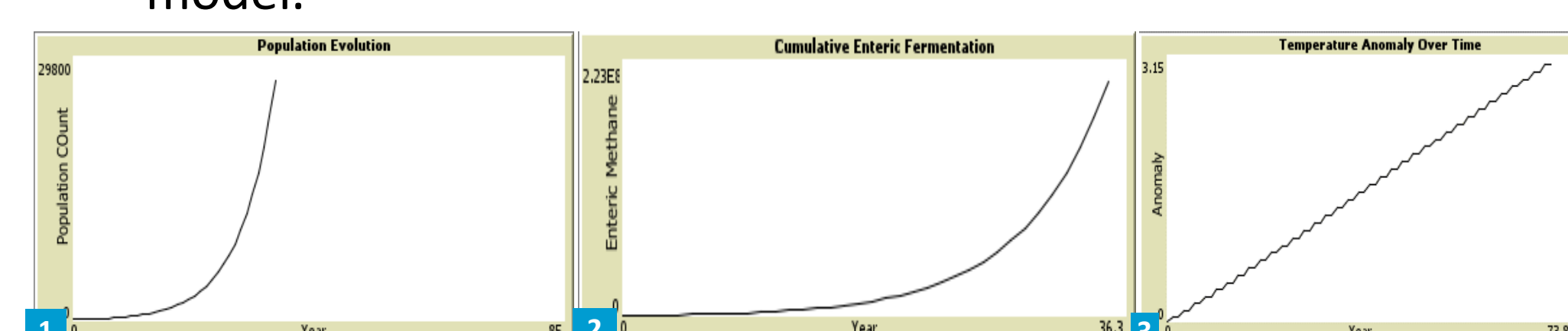
Diet Quality	SSP 1 – 1.9	SSP 1 – 2.6	SSP 2 – 4.5	SSP 3 – 7.0	SSP 5 – 8.5
High Quality	0.075 ± 0.0007	0.14 ± 0.0013	0.197 ± 0.0018	0.28 ± 0.0028	0.356 ± 0.004
Moderate Quality	0.081 ± 0.0007	0.152 ± 0.0013	0.212 ± 0.002	0.31 ± 0.003	0.383 ± 0.004
Low Quality	0.083 ± 0.0007	0.154 ± 0.0013	0.216 ± 0.002	0.315 ± 0.003	0.389 ± 0.004

- 1) Temperature anomaly (in C) under different SSP scenarios and qualities of diet generated using NetLogo 6.4.0 software upon running 1240 simulations.

Modelling Temperature Anomaly due to Enteric Methane Emissions.

The study first models the population, followed by modelling enteric emissions emitted under parameters – quality of diet, number of hours worked and net energy taken up by each cattle to perform different activities, and uses these emissions as CO₂ eq to estimate temperature anomaly under SSP scenarios. This approach establishes the first link between CH₄ emissions influencing the climate.

- The population is modelled at 60% reproductive rate and 15% mortality rate with the lifespan of 16 years and the purpose of dairy.
- This model projects the vulnerability caused due to cattle on the climate via temperature anomaly over 35 years.
- The bifurcation while modelling for enteric emission enables us to understand the net energy taken up by cattle and design policy prescriptions at a microscopic level.
- The feeding style – pasture, grazing and stall – have also been taken into consideration. Cattle that were fed via pastures emitted less methane compared to stall and grazing.
- The correlation between temperature anomaly under SSP scenarios and number of hours worked is 0.04. While the correlation is moderately strong, it indicates that over working the cattle increase its net energy for work and activity; therefore, increases the amount of CH₄ it emits.
- **Limitations:** The model assumes uniform diet, working hours, and general net energy across the entire population, limiting accuracy. No externalities are accounted for in the model.



- 1) Population model – first component of model.
- 2) Cumulative enteric methane emissions – second component of model.
- 3) Temperature anomaly due CO₂ eq of CH₄ emissions under SSP 5-8.5 – third component of model.

Recommendations.

Feeding Strategies:

- Provision of high quality diet with increased concentrates (concentrates >90).
- Provision of pasture feeds and rotational grazing.
- Increasing protein supplement in the diet.
- Using the technique of precision feeding in order to administer proportionate diets.
- Improve forage quality. High sugar ryegrass leads to a 7.5–21.0% increase in milk yield and a 7.1–25.7% decrease in excrement nitrogen. Re-seeding native grass species with those with higher productivity or C allocation to deeper roots, or introducing legumes into grazing lands, can all promote soil C in rangeland soils and reduce N emissions[1].
- Dietary inclusion of free oils or oil-rich feeds (e.g. oilseeds, distillers' grains, and micro-algae), biologically active plant compounds (e.g. condensed tannins, saponins, and essential oils), rumen fermentation modifiers (e.g. yeast and bacterial direct-fed microbials)[5].

Rumen Manipulation Strategies:

- Use of bacteriocins, ionophores, fats/oils, prebiotics and probiotics, halogenated methane analogues[1].

Population Modelling Strategies:

- Optimal livestock rearing.
- Improving breed management.
- Improving manure management.
- Administering methanogens vaccines to optimise VFAs.
- Regular health checkups.

General Recommendations :

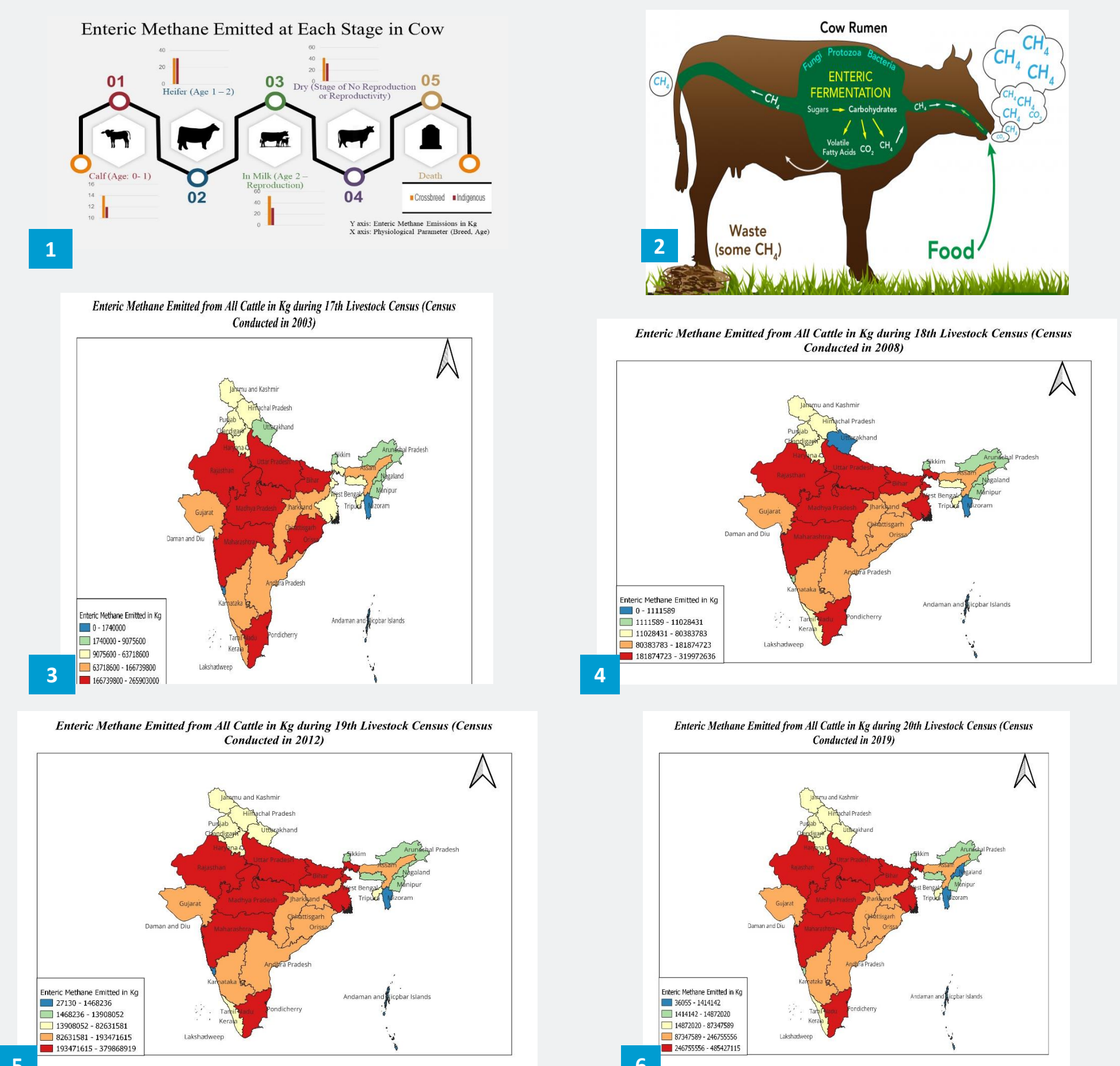
- Creation of more intergovernmental policies and programs for creating mitigation strategies.

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- 3) Credits to Dr. Miguel Rodriguez at Universität Hamburg.



- 1) Enteric methane emitted at different stages of life of a cow [3].
- 2) Process of enteric fermentation in cattle (Photo: Let's Talk Science).
- 3) Enteric methane emissions of cattle calculated using 17th livestock census.
- 4) Enteric methane emissions of cattle calculated using 18th livestock census.
- 5) Enteric methane emissions of cattle calculated using 19th livestock census.
- 6) Enteric methane emissions of cattle calculated using 20th livestock census